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$\mathbb{Z}_2$ lattice gauge theory on non-trivial topology and its quantum simulation

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Abstract: Wegner duality is essential for $\mathbb{Z}_2$ lattice gauge theory, yet the duality on non-trivial topologies has remained implicit. We extend Wegner duality to arbitrary topology and dimension, obtaining a new class of Ising models, in which topology is encoded in non-local domain-wall patterns. Without the overhead of gauge constraints, simulating this model on an $L \times L$ torus requires only L^2 qubits with two-body couplings, halving the conventional four-body coupled $2L^2$ qubits, enabling full experimental realization of $\mathbb{Z}_2$ lattice gauge theory on near-term devices.

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1 Introduction

Lattice gauge theories (LGTs) originate from Wilson's non-perturbative framework for quantum chromodynamics [1] and the subsequent Hamiltonian formulation by Kogut and Susskind [2]. Duality of LGTs [3] was introduced by Wegner, with the map between (2+1)D $\mathbb{Z}_2$ lattice gauge theory ($\mathbb{Z}_2$ LGT) and the Ising model [4], later generalized to $\mathbb{Z}_N$, U(1) [5] and SU(N) [6] LGTs. The physical significance of $\mathbb{Z}_2$ LGT was further elevated by topological orders in a variety of systems such as fractional quantum Hall effect [7], string-net condensation [8–11], superconductivity [12], and quantum error correction codes, e.g. the toric code [13].

This trend has stimulated considerable recent experimental activity in the quantum simulation of $\mathbb{Z}_2$ LGTs demonstrated on both classical emulators [14–19] and on near-term quantum processing platforms [20–26]. In a complementary direction, the original Wegner duality has been applied inversely to map challenging optimization problems, encoded as Ising models, onto experimentally more accessible $\mathbb{Z}_2$ LGT Hamiltonians for quantum annealing [27] and quantum inspired classical optimization algorithms [28].

Wegner's duality indicates a promising approach for quantum simulation of $\mathbb{Z}_2$LGTs [29], i.e. simulating the dual Ising model instead of directly dealing with $\mathbb{Z}_2$LGT, which possesses high-order couplings and gauge constraints. However, the most discussed $\mathbb{Z}_2$LGT on a toroidal geometry exhibits fourfold degeneracy in its ground states, a characteristic absent in its dual Ising model. This distinction arises because Wegner's duality is exact on simply connected

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spaces. Its extension to non-trivial topologies has remained implicit for quantum simulation. Although pioneering works [3, 5, 30] have discussed $\mathbb{Z}_2$LGT on the torus, there has been a lack of explicit spin model for the Wegner duality of $\mathbb{Z}_2$LGT on non-trivial topologies, forming a major obstacle for experimental studies.

In this paper, we extend the Hamiltonian form of Wegner duality and its use in quantum simulation to non-trivial topologies that are compatible with near-term intermediate scale quantum (NISQ) devices. Using a homological framework similar to CSS quantum error correction codes [13, 31, 32], we show that local dynamics of the $\mathbb{Z}_2$ LGT is dominated by fluctuations of closed strings, while the ground state degeneracy corresponds to the homology group. In the dual model, spins correspond to local closed strings in the original model, and there are nonlocal couplings with the auxiliary spins, different values of which correspond to different topological sectors. This gives rise to a *Sectorial Ising (SI) model*, which enables gauge-free quantum simulation of an $L \times L$ torus with L^2 qubits per sector, halving the $2L^2$ qubit requirement in conventional approaches at the cost of multiple independent runs. Since SI model requires fewer qubits than the toric code model, it stands as a new resource-efficient model for quantum simulation of topological order.

2 Closed string representation of $\mathbb{Z}_2$LGT

Let Σ, Λ and V be the plaquettes, links and vertices of an $L \times L$ torus, and on each link ℓ we place a qubit, with Pauli operators for x and z directions denoted as X_ℓ and Z_ℓ, respectively. The star and plaquette operators detect $\mathbb{Z}_2$ electric charge and magnetic vortices, respectively

$$A_v = \prod_{\ell \in \delta v} X_\ell, \quad B_\sigma = \prod_{\ell \in \partial \sigma} Z_\ell, \tag{2.1}$$

where $\partial\sigma \subset \Lambda$ are the boundary links of plaquette σ, and $\delta v \subset \Lambda$ are the links adjacent to vertex v. $\mathbb{Z}_2$LGT is formulated as

$$H = -\sum_{\sigma \in \Sigma} \prod_{\ell \in \partial \sigma} Z_\ell + g \sum_{\ell \in \Lambda} X_\ell, \tag{2.2}$$

of which $\{A_v\}_{v \in V}$ is the gauge symmetry operators.

The states of the $\mathbb{Z}_2$LGT are in one-to-one correspondence with superpositions of closed strings, giving rise to a *closed string representation*. We choose gauge fixing as the $+1$ common eigenspace of the $\{A_v\}$ operators. The closed strings are excited by Wilson string operators $\prod_{\ell \in w} Z_\ell$ for each loop w on the primal lattice, forming a basis of the gauge-fixed Hilbert space, as proved in appendix A. The deconfined phase ($g = 0$) ground state of $\mathbb{Z}_2$LGT in the trivial topological sector, $|g_{00}\rangle \sim \sum_w \prod_{\ell \in w} Z_\ell |+\rangle$, is the superposition of all contractible closed strings, an example of Wen's string-net condensation [10]. States in non-trivial topological sectors are obtained by $\tilde{X}_\gamma = \prod_{\ell \in \gamma} X_\ell, \gamma = \gamma_1, \gamma_2$ being the non-contractible loops on the torus.

3 Wegner duality on non-trivial topology

The conventional Wegner duality, in absence of topological non-triviality, maps the plaquette operator B_σ to the Pauli operator $\tilde{X}_\sigma$ of a qubit on plaquette σ, and X_ℓ to $\prod_{\sigma \in \delta \ell} \tilde{Z}_\ell$ where

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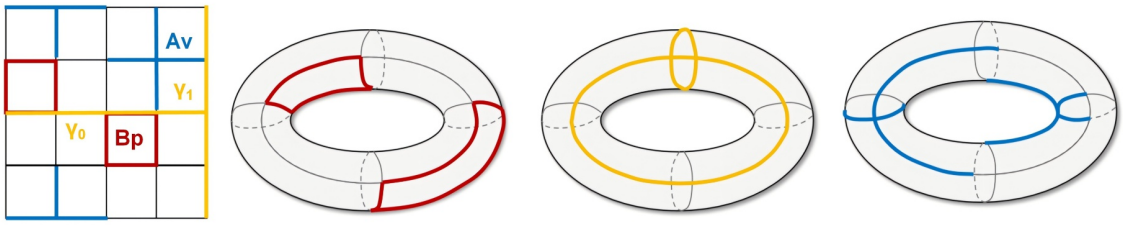


Figure 1. Hodge decomposition of the string Hilbert space into contractible closed strings (red), harmonic fields (gold), and gauge transformations (blue), mathematically described by $\mathrm{im}(\partial)$, $\ker(\Delta)$, and $\mathrm{im}(\delta)$.

$\delta\ell \subset \Lambda$ are the plaquettes adjacent to ℓ. In case of non-trivial topology, we place a qubit not only on each plaquette, but also on each independent non-contractible loop, e.g. γ_1, γ_2 on the torus. Then we transform to the closed string representation by

$$\tilde{X}_\sigma = \prod_{\ell \in \partial\sigma} Z_\ell, \tag{3.1}$$

$$X_\ell = \Big(\prod_{\sigma \in \delta\ell} \tilde{Z}_\sigma\Big)\Big(\prod_{\gamma \ni \ell} \tilde{Z}_\gamma\Big). \tag{3.2}$$

This formulation extends the Wegner duality with the topological term $\tilde{Z}_\gamma$ corresponding to non-trivial loop γ entering the link ℓ.

4 General topology and the Hodge perspective

The aforementioned transformation is extended to general topologies in arbitrary dimension. To capture the commuting algebra of the star (A_v) and plaquette (B_p) operators in $\mathbb{Z}_2$LGT, we employ combinatorial Hodge theory, natural frameworks to decompose a field into source-free, curl-free and harmonic components.

Consider a general cell complex (e.g. a lattice)

$$\mathbb{Z}_2^\Sigma \underset{\delta}{\overset{\partial}{\rightleftharpoons}} \mathbb{Z}_2^\Lambda \underset{\delta}{\overset{\partial}{\rightleftharpoons}} \mathbb{Z}_2^V, \quad \partial^2 = 0, \delta^2 = 0. \tag{4.1}$$

where V, Λ, and Σ are three finite sets, ∂ and δ being the boundary and coboundary operators with relation $\delta = \partial^T$. Here we implicitly assume the isomorphism between chain groups and cochain groups, which allows the boundary operator ∂ and the coboundary operator δ to act in the same spaces $\mathbb{Z}_2^\Sigma$, $\mathbb{Z}_2^\Lambda$, and $\mathbb{Z}_2^V$. For the mathematical construction that explicitly handles this subtlety, please refer to appendix B.

On each element $\ell \in \Lambda$ resides a spin, with all possible spin configurations (up/down) described by $\mathbb{Z}_2^\Lambda$. At each element $v \in V$ there can be a $\mathbb{Z}_2$ electric charge, with all charge configurations described by $\mathbb{Z}_2^V$. The star operators $\{A_v\}$ to detect these charges are described by the coboundary operator δ. At each element $\sigma \in \Sigma$ there can be a $\mathbb{Z}_2$ magnetic flux, with all flux configurations described by $\mathbb{Z}_2^\Sigma$. The plaquette operators $\{B_p\}$ to detect these fluxes are described by the boundary operator ∂.

On non-trivial topologies, star and plaquette operators — encoding divergence and curl of $\mathbb{Z}_2$ LGT fields — cannot uniquely determine the field (unlike Euclidean space). The

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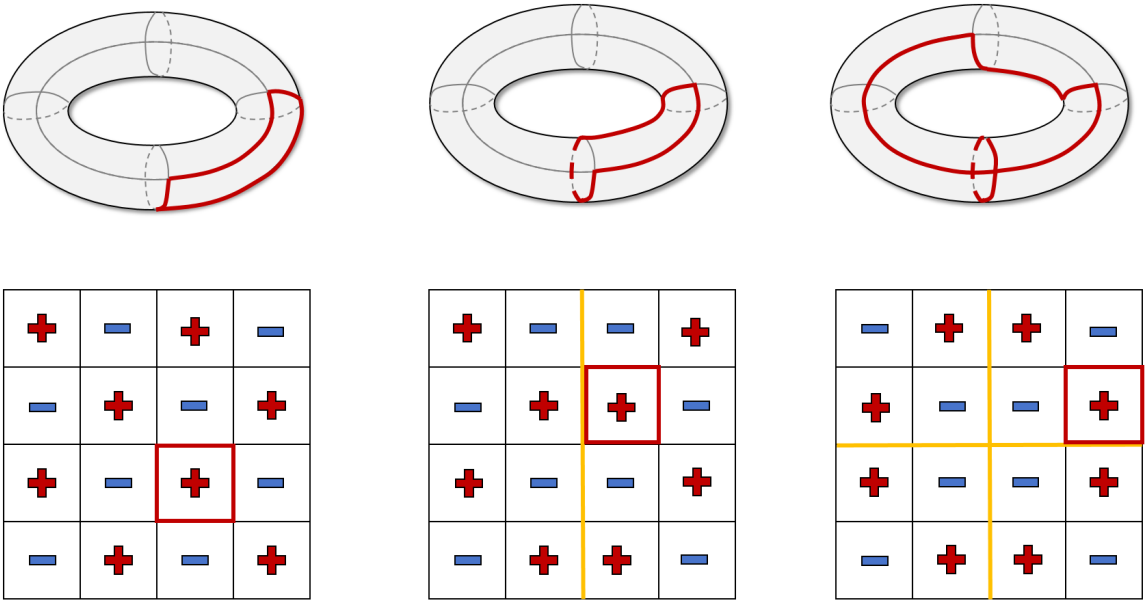


Figure 2. Ground state of SI model at $g = 0$: each sector (left/mid/right) has a unique ground state set by domain walls; left column with trivial topology reduces to the Ising model. Dual spin (red border) maps to a closed string in the corresponding $\mathbb{Z}_2$LGT sector.

missing harmonic component is captured by the Hodge-Laplace operator $\Delta = \partial\delta + \delta\partial$, defining $\mathcal{T} = \ker\Delta$ as the subspace that is both curl- and source-free, encoding purely topological properties. The combinatorial Hodge theorem gives an orthogonal decomposition of the Hilbert space:

$$\mathbb{Z}_2^{\Lambda} = \mathrm{im}(\delta) \oplus \mathcal{T} \oplus \mathrm{im}(\partial). \tag{4.2}$$

This decomposition separates the physical degrees of freedom: $\mathrm{im}(\delta)$ constitutes the gauge degrees of freedom of the $\mathbb{Z}_2$LGT, $\mathcal{T}$ comprises harmonic fields with a one-to-one mapping to the topological sectors, responsible for the ground state degeneracy, and $\mathrm{im}(\partial)$ encodes the dynamics of contractible strings, which are local degrees of freedom.

By projecting the $\mathbb{Z}_2$LGT Hamiltonian to the gauge-invariant subspace $\mathcal{T} \oplus \mathrm{im}(\partial)$ via the duality transformation (3.1), (3.2), we define the gauge-free *sectorial Ising (SI) model*.

5 Sectorial Ising model

Applying the dual transformation (3.1) and (3.2), we obtain the Hamiltonian dual to $\mathbb{Z}_2$LGT on non-trivial topology

$$H' = -\sum_{\sigma\in\Sigma} \tilde{X}_\sigma + g \sum_{\ell\in\Lambda} \prod_{\sigma\in\delta\ell} \tilde{Z}_\sigma \cdot \prod_{\gamma\ni\ell} \tilde{Z}_\gamma. \tag{5.1}$$

This Hamiltonian takes the transverse-field Ising form, but with nonlocal couplings to auxiliary qubits $\{\tilde{Z}_\gamma\}$ that encode the topological sector. We refer to this as the Sectorial Ising (SI) model, where the dual of a non-contractible loop in the $\mathbb{Z}_2$LGT is an auxiliary topological qubit. In the trivial sector ($\tilde{Z}_\gamma = +1$), it reduces to the standard Ising model.

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Note that the plaquette operators in the original $\mathbb{Z}_2$LGT are dependent. States with $\prod_{\sigma\in\Sigma}\tilde{X}_\sigma = +1$ symmetry describes the dynamics of the $\mathbb{Z}_2$LGT. As $[\prod_{\sigma\in\Sigma}\tilde{X}_\sigma, H'] = 0$, it suffices to ensure the symmetry of the initial state to be $\prod_{\sigma\in\Sigma}\tilde{X}_\sigma = +1$.

The redundancy in the plaquette operators reveals topology characterized by non-contractible surfaces other than non-contractible loops. It can be eliminated by depicting a plaquette σ_0 and let $\Sigma^* = \Sigma - \{\sigma_0\}$, establishing

$$H' = -H'_M + gH'_E, \tag{5.2}$$

$$H'_M = \sum_{\sigma\in\Sigma^*}\tilde{X}_\sigma + \prod_{\sigma\in\Sigma^*}\tilde{X}_\sigma, \tag{5.3}$$

$$H'_E = \sum_{\ell\in\Lambda}\prod_{\sigma\in\delta\ell\cap\Sigma^*}\tilde{Z}_\sigma \cdot \prod_{\gamma\ni\ell}\tilde{Z}_\gamma. \tag{5.4}$$

Origin of the global product term in (5.3) is mathematically described by the Betti number and high-order cohomology groups, as shown in appendix B.

6 The sectorial Ising model as an emergent topological phenomena

The SI model has two key features: (1) it describes the same wavefunction as $\mathbb{Z}_2$ gauge field theory through a topologically induced construction, and (2) it shares features like degenerate ground states with topological materials, but crucially differs in its handling of sector transitions.

Remarkably, the operators $\tilde{Z}_\gamma$ commute with (5.1). This inherent symmetry allows further reduction of $\mathcal{T}$ in (4.2), leaving only $\mathrm{im}(\partial)$ degrees of freedom. This yields the Hamiltonian

$$H'(\{z_\gamma\}) = -\sum_{\sigma\in\Sigma}\tilde{X}_\sigma + g\sum_{\ell\in\Lambda}\prod_{\gamma\ni\ell}(-1)^{(1-z_\gamma)/2}\prod_{\sigma\in\delta\ell}\tilde{Z}_\sigma. \tag{6.1}$$

Trivial sector. When $\tilde{Z}_\gamma = +1$ for all $\gamma \in \mathcal{C}$, the Hamiltonian reduces to a transverse-field Ising model. This essentially recovers the result given by Wegner, Fradkin, and Susskind.

Non-trivial sectors. For configurations where $\tilde{Z}_{\gamma_i} = -1$ on a set of non-contractible loops $\{\gamma_i\}$, the Hamiltonian describes a phase transition from a ferromagnetic state to a paramagnetic state with domain walls, which constitutes a generalized Ising model. Each distinct configuration of $\{\tilde{Z}_\gamma = \pm 1\}$ defines a unique topological sector of the $\mathbb{Z}_2$ gauge theory.

In the ferromagnetic phase, the dominant energy contribution comes from the magnetic term, rendering the sign of the electric term physically irrelevant. Consequently, SI model exhibits identical behavior to the Ising model under ferromagnetic ordering. This demonstrates how topologically protected phenomena in the $\mathbb{Z}_2$ gauge theory manifest in its dual Ising model.

The SI model possesses an intrinsic gauge freedom: for a fixed configuration of $\{\tilde{Z}_\gamma = \pm 1\}$, non-contractible loops can be selected in numerous distinct ways. Each choice yields a Hamiltonian with different coefficients, yet all share identical Hilbert spaces — analogous to a group admitting multiple generator choices. This freedom can be experimentally advantageous, allowing selection of higher-quality qubits to form the non-contractible loops and enhance fidelity.

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The SI model eliminates gauge degrees of freedom but preserves closed string dynamics of the $\mathbb{Z}_2$LGT, maintaining string-net condensation with topological order and sector classification. While the toric code was historically deemed the simplest such model — typically requiring $2L^2$ qubits on an $L \times L$ lattice — the SI model achieves the same physics with up to three-body couplings at the cost of gauge fixing, fewer than the four-body terms required for the toric code.

Since a transverse Ising model can be mapped to a higher-dimensional classical Ising model, the SI model can similarly correspond to the following model:

$$H'_{csi}(\{z_\gamma\}) = -K \sum_{\sigma\in\Sigma} \left(s_{\sigma,1} s_{\sigma,N} + \sum_{n=1}^{N-1} s_{\sigma,n} s_{\sigma,n+1} \right) + g\Delta\tau \sum_{n=1}^{N} \sum_{\ell\in\Lambda} \prod_{\gamma\ni\ell} (-1)^{(1-z_\gamma)/2} \prod_{\sigma\in\delta\ell} s_{\sigma,n}. \tag{6.2}$$

where imaginary time direction has been divided into N time slices of width $\Delta\tau$ and $\exp(-2K) = \tanh(\Delta\tau)$ [33]. Using techniques such as Monte Carlo methods for simulating Ising models, we can deal with larger lattice scales on current computer hardware and also provide guidance for future simulations on quantum computers.

7 Examples of $\mathbb{Z}_2$LGT with non-trivial topology

Topological structures are essential for the performance of topological quantum codes. Similarly, by employing the language of chain complexes, we can characterize the $\mathbb{Z}_2$LGT Hamiltonian and its dual on arbitrary topology. As shown in the following examples, however, not all non-trivial topological properties(e.g., non-zero Betti numbers in examples 1 and 3) provide protection.

Example 1. For $\mathbb{Z}_2$LGT on a tetrahedron as shown in figure 3(a),

$$H_M = Z_1Z_2Z_4 + Z_2Z_3Z_5 + Z_1Z_3Z_6 + Z_4Z_5Z_6,$$
$$H_E = X_1 + X_2 + X_3 + X_4 + X_5 + X_6,$$

with gauge symmetry operators $X_1X_2X_3$, $X_1X_4X_6$, $X_2X_4X_5$, and $X_3X_5X_6$. The dual Hamiltonian is

$$H'_M = \tilde{X}_1 + \tilde{X}_2 + \tilde{X}_3 + \tilde{X}_1\tilde{X}_2\tilde{X}_3,$$
$$H'_E = \tilde{Z}_1\tilde{Z}_2 + \tilde{Z}_2\tilde{Z}_3 + \tilde{Z}_1\tilde{Z}_3 + \tilde{Z}_1 + \tilde{Z}_2 + \tilde{Z}_3.$$

Note that if $Z_4Z_5Z_6$ is removed from H_M, the global product term $\tilde{X}_1\tilde{X}_2\tilde{X}_3$ in H'_M should also vanish since the H_M topology becomes the disk, and the dual Hamiltonian also becomes the Ising model. From the mesh perspective, this phenomenon describes a bulk-edge effect. By Stokes theorem

$$\prod_{\ell\in\partial\sigma_0} Z_\ell = \prod_{\sigma\in\Sigma^*} \prod_{\ell\in\partial\sigma} Z_\ell,$$

the right hand side is $\prod_{\sigma\in\Sigma^*} \tilde{X}_\sigma$. The global product term describes $\mathbb{Z}_2$ magnetic flux living on the boundary of the open surface Σ^*. The total $\mathbb{Z}_2$ magnetic flux is conserved on the surface and its boundary.

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Example 2. $\mathbb{Z}_2$LGT on a 2×2 torus, as shown in figure 3(b), has the Hamiltonian

$$\begin{aligned} H_M &= Z_{000}Z_{001}Z_{011}Z_{100} + Z_{010}Z_{011}Z_{001}Z_{110} \\ &\quad + Z_{100}Z_{101}Z_{111}Z_{000} + Z_{110}Z_{111}Z_{101}Z_{010}, \\ H_E &= X_{000} + X_{010} + X_{100} + X_{110} \\ &\quad + X_{001} + X_{101} + X_{011} + X_{111}. \end{aligned}$$

and gauge symmetry operators $X_{000}X_{001}X_{101}X_{101}$, $X_{000}X_{010}X_{011}X_{111}$, $X_{001}X_{101}X_{100}X_{110}$, and $X_{011}X_{111}X_{100}X_{110}$. The dual Hamiltonian is

$$\begin{aligned} H'_M &= \tilde{X}_{10} + \tilde{X}_{01} + \tilde{X}_{11} + \tilde{X}_{01}\tilde{X}_{10}\tilde{X}_{11}, \\ H'_E &= \tilde{Z}_{\gamma_0}\tilde{Z}_{10} + \tilde{Z}_{\gamma_0}\tilde{Z}_{01}\tilde{Z}_{11} + \tilde{Z}_{10} + \tilde{Z}_{01}\tilde{Z}_{11} \\ &\quad + \tilde{Z}_{\gamma_1}\tilde{Z}_{01} + \tilde{Z}_{\gamma_1}\tilde{Z}_{10}\tilde{Z}_{11} + \tilde{Z}_{01} + \tilde{Z}_{10}\tilde{Z}_{11}. \end{aligned}$$

The torus has non-trivial fundamental group whose generators γ_0, γ_1 are explicit in the closed string Hamiltonian.

Example 3. Consider two tetrahedrons sharing a common link, as shown in figure 3(c). $\mathbb{Z}_2$LGT on such geometry has the Hamiltonian

$$\begin{aligned} H_M &= Z_1Z_2Z_6 + Z_1Z_3Z_5 + Z_2Z_3Z_4 + Z_4Z_5Z_6 \\ &\quad + Z_6Z_7Z_8 + Z_6Z_{10}Z_{11} + Z_7Z_9Z_{10} + Z_8Z_9Z_{11}, \\ H_E &= \sum_{i=1}^{11} X_i. \end{aligned}$$

and gauge symmetry operators $X_1X_2X_3$, $X_3X_4X_5$, $X_1X_5X_7X_{10}$, $X_2X_4X_8X_{11}$, $X_4X_5X_6$, and $X_9X_{10}X_{11}$. The dual Hamiltonian is

$$\begin{aligned} H'_M &= \tilde{X}_1 + \tilde{X}_2 + \tilde{X}_3 + \tilde{X}_1\tilde{X}_2\tilde{X}_3 + \tilde{X}_4 + \tilde{X}_5 + \tilde{X}_6 + \tilde{X}_4\tilde{X}_5\tilde{X}_6, \\ H'_E &= \tilde{Z}_1\tilde{Z}_2 + \tilde{Z}_1\tilde{Z}_3 + \tilde{Z}_2\tilde{Z}_3 + \tilde{Z}_2 + \tilde{Z}_3 + \tilde{Z}_1\tilde{Z}_4 \\ &\quad + \tilde{Z}_4\tilde{Z}_5 + \tilde{Z}_5\tilde{Z}_6 + \tilde{Z}_4\tilde{Z}_6 + \tilde{Z}_5 + \tilde{Z}_6. \end{aligned}$$

This example shows that the topological properties of a Hamiltonian can be different from the geometry that defines the system, as the mesh in figure 3(c) is planar, but the Hamiltonian has the topology of a two tetrahedron union.

8 Digital simulation of $\mathbb{Z}_2$LGT on a torus

The duality between $\mathbb{Z}_2$LGT and sectorial Ising models provides a gauge-free model for quantum simulation. A $\mathbb{Z}_2$LGT model can be simulated with minimal resources by performing independent quantum simulations for each sector of the sectorial Ising model. Thus $\mathbb{Z}_2$LGT on the $L \times L$ torus can be simulated by L^2 qubits per sector, requiring only two-body couplings.

Quantum phase transition of $\mathbb{Z}_2$LGT on torus of different sizes is observed, as shown in figure 4. The critical point g_c is measured as extrema points of $\mathrm{d}\langle x\rangle/\mathrm{d}g$ and $\mathrm{d}\langle z\rangle/\mathrm{d}g$, which are obtained by central difference of step 0.01 and moving average filter of window size

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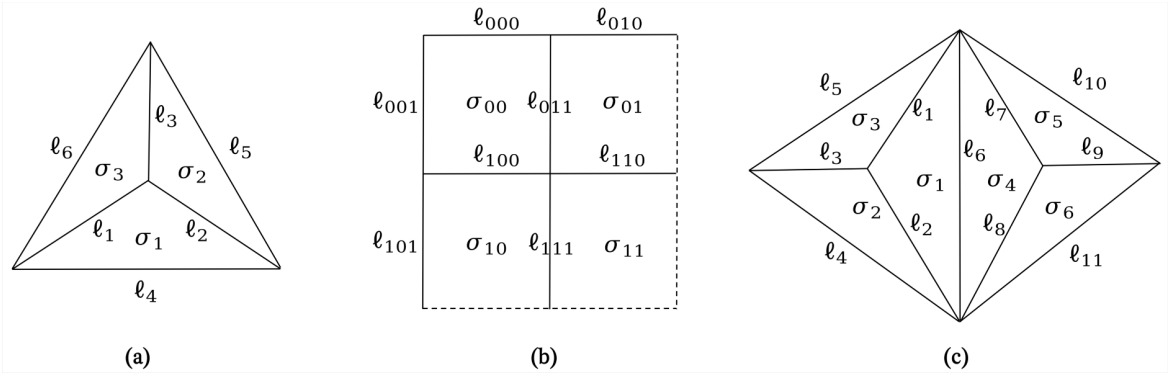


Figure 3. Geometric configurations of Example 1(a), 2(b) and 3(c).

0.01, showing $g_c \approx 0.36$ on 5×5 torus and increases when lattice size decreases. The energy gap ΔE between ground states $|g_{00}\rangle$ and $|g_{01}\rangle$ agrees with $\Delta E \propto g^{\frac{1}{L+1}}$ near the origin [34]. The expectation of Wilson loops obeys perimeter law and area law in the deconfined and confined phases respectively. The simulation results are consistent with the theoretical prediction of $\mathbb{Z}_2$LGT [34].

9 Conclusion

We extend Wegner duality of the $\mathbb{Z}_2$LGT to non-trivial topology, formulating the sectorial Ising (SI) model together with its classical form, that is highly compatible for near-term quantum simulators. Topological protection emerges naturally from the correspondence between topological sectors of the $\mathbb{Z}_2$LGT and domain-wall configurations of the SI model. This model enables resource-efficient quantum simulation of $\mathbb{Z}_2$LGTs, reducing the simulation cost of $\mathbb{Z}_2$LGT on an $L \times L$ torus from four-body coupled $2L^2$ qubits to the two-body coupled L^2 qubits of the sectorial Ising model.

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A Completeness of closed string representation proved by Euler's formula

We show that all states of the $\mathbb{Z}_2$LGT within the gauge fixed $\{A_v = +1\}_{v \in V}$ subspace $\mathcal{H}_G$ are superpositions of closed strings. We show with explicit construction that (i) the closed strings are in $\mathcal{H}_G$, and (ii) $\mathcal{H}_G$ contains only the closed strings due to the constraint of dimensionality.

Note that $|+\rangle \in \mathcal{H}_G$. Moreover, any loop w intersects even number of links adjacent to a vertex,

$$\left[\prod_{\ell \in w} Z_\ell, A_v\right] = 0.$$

Thus $\prod_{\ell \in w} Z_\ell |+\rangle \in \mathcal{H}_G$, proving (i).

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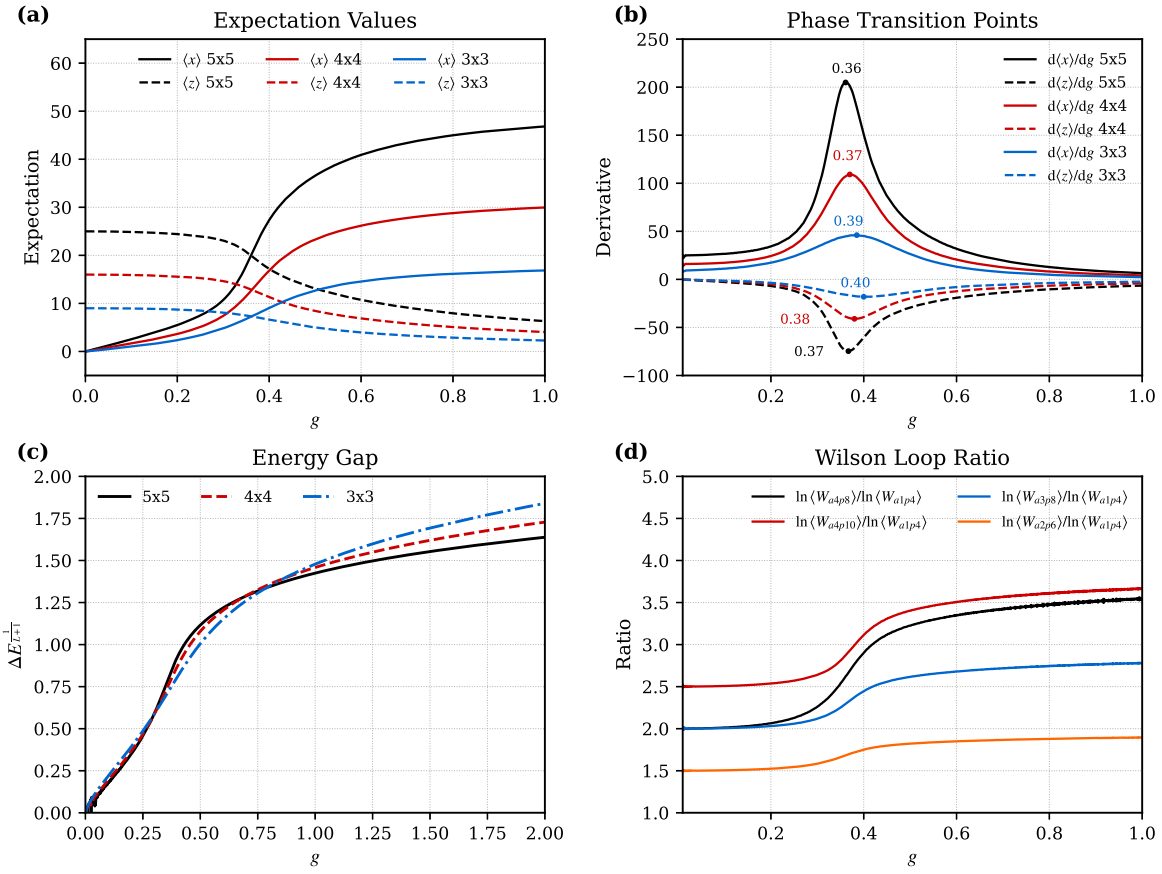


Figure 4. Quantum phase transitions in the model (5.2)–(5.4) on $L \times L$ tori. (a) Energy expectations. (b) Critical points g_c. (c) Gap ΔE. (d) Wilson loops $\langle W_C \rangle$. These results, fully consistent with direct calculations of $\mathbb{Z}_2$LGT, correspond to figures 8, 9, 16, and 7 in our earlier work [14], respectively. 100,000 Trotter steps with step-size 0.01 controls $\langle x \rangle = \langle H_E \rangle$ and $\langle z \rangle = \langle H_M \rangle$ absolute errors below 10^{-3}. Gradient is calculated by central differential method of step 0.01, with moving average filter window 0.01.

On a closed surface S with g_S genus, there are $|\Sigma| - 1$ independent plaquette borders due to closed topology of S, and there are two independent non-contractible loops across each genus, rendering

$$\dim(\mathcal{H}_G) \geq 2^{|\Sigma|+2g_S-1}.$$

On the other hand, a closed surface has $|\Lambda|$ qubits and $|V| - 1$ independent gauge operators, demanding

$$\dim(\mathcal{H}_G) \leq 2^{|\Lambda|-|V|+1}.$$

By Euler's formula $|\Sigma| - |\Lambda| + |V| = 2 - 2g_S$, we show $\dim(\mathcal{H}_G) = 2^{|\Sigma|+2g_S-1}$, proving (ii).

B Homological structure of 2 times 2 torus

In this appendix we investigate the homological structure of the 2×2 torus as shown in figure 5. Let the set of all plaquettes, links and vertices be $\Sigma = \{\sigma_{00}, \sigma_{01}, \sigma_{10}, \sigma_{11}\}$,

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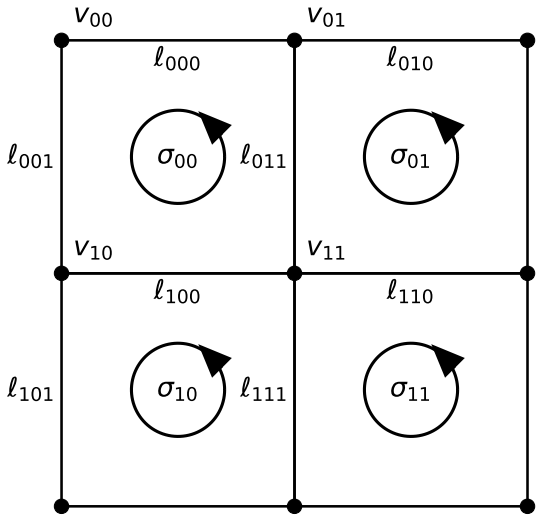


Figure 5. A 2×2 lattice with periodic boundary condition.

$\Lambda = \{\ell_{000}, \ell_{001}, \ell_{010}, \ell_{011}, \ell_{100}, \ell_{101}, \ell_{110}, \ell_{111}\}$ and $V = \{v_{00}, v_{01}, v_{10}, v_{11}\}$. We define chain groups as the free vector space over $\mathbb{Z}_2$,

$$C_0 = \sum_{v \in V} c_v v, \quad C_1 = \sum_{\ell \in \Lambda} c_\ell \ell, \quad C_2 = \sum_{\sigma \in \Sigma} c_\sigma \sigma,$$

where $c_v, c_\ell, c_\sigma \in \mathbb{Z}_2$. They are organized by the boundary operators ∂_2, ∂_1, or ∂ when determined by context,

$$\partial_2 \sigma = \sum_{\ell \text{ is boundary of } \sigma} \ell,$$
$$\partial_1 \ell = \sum_{v \text{ is boundary of } \ell} v.$$

e.g. $\partial \sigma_{00} = \ell_{000} + \ell_{001} + \ell_{100} + \ell_{011}$, $\partial \ell_{000} = v_{00} + v_{01}$. An important property of the boundary operator is

$$\partial \circ \partial = 0,$$

e.g.

$$\partial_1 \circ \partial_2(\sigma_{00}) = \partial_1(\ell_{000}+\ell_{001}+\ell_{100}+\ell_{011}) = (v_{00}+v_{01})+(v_{01}+v_{11})+(v_{10}+v_{11})+(v_{00}+v_{10}) = 0.$$

In other words, the border of some plaquettes always intersects even number of links adjacent to a vertex. This establishes the chain complex

$$C_2 \xrightarrow{\partial_2} C_1 \xrightarrow{\partial_1} C_0, \quad \partial_1 \circ \partial_2 = 0.$$

The topology is well characterized by the chain complex. The boundary group B_i and the cycle group Z_i are defined as

$$B_i = \mathrm{im}(\partial_{i+1}),$$
$$Z_i = \ker(\partial_i).$$

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In our 2×2 torus example,

$$B_1 = \langle \partial\sigma_{10}, \partial\sigma_{01}, \partial\sigma_{11} \rangle$$
$$Z_1 = \langle \gamma_0, \gamma_1, \partial\sigma_{10}, \partial\sigma_{01}, \partial\sigma_{11} \rangle,$$

where $\gamma_0 = \ell_{000} + \ell_{010}$ and $\gamma_1 = \ell_{001} + \ell_{101}$. The boundary group is literally generated by plaquettes boundaries, and the cycle group consists of chains without boundaries.

Due to $\partial \circ \partial = 0$ we always have $B_1 \subset Z_1$, geometrically showing that a boundary is always a cycle. However, a cycle is not necessarily a boundary, e.g. $\gamma_0, \gamma_1 \notin B_1$. The difference is characterized by the homology group

$$H_i = Z_i / B_i,$$

In our 2×2 torus example, $H_1 = \langle \gamma_0, \gamma_1 \rangle$.

The rank of homology groups are Betti numbers,

$$b_i = \dim H_i.$$

For $\mathbb{Z}_2$LGT on the torus, $b_2 = 1$ is the number of dependent plaquette operators and $b_1 = 2$ is the number of topological qubits.

Cohomology emerges naturally when considering $\mathbb{Z}_2$ valued functions over chain groups, the cochain groups

$$C^i = \mathrm{Hom}(C_i, \mathbb{Z}_2).$$

The cochain groups are generated by characteristic functions over the chain groups, e.g. $C^2 = \langle \mathbf{1}_\sigma \rangle_{\sigma \in \Sigma}$, $C^1 = \langle \mathbf{1}_\ell \rangle_{\ell \in \Lambda}$, and $C^0 = \langle \mathbf{1}_v \rangle_{v \in V}$, where the characteristic functions $\mathbf{1}_\sigma$ are defined by Kronecker pairing

$$\langle \mathbf{1}_\sigma, \tau \rangle = \mathbf{1}_\sigma(\tau) = \delta_{\sigma\tau}.$$

Cochain groups are organized by coboundary operators δ^0, δ^1, or δ when the context is clear,

$$\delta^0 \mathbf{1}_v = \sum_{\ell : v \in \partial\ell} \mathbf{1}_\ell,$$
$$\delta^1 \mathbf{1}_\ell = \sum_{\sigma : \ell \in \partial\sigma} \mathbf{1}_\sigma,$$

giving rise to the cochain complex

$$C^0 \xrightarrow{\delta^0} C^1 \xrightarrow{\delta^1} C^2, \quad \delta^1 \circ \delta^0 = 0.$$

The cohomology group is

$$H^i = Z^i / B^i = \ker(\delta^i) / \mathrm{im}(\delta^{i-1}).$$

The cohomology group for our 2×2 torus is $H^1 = \langle \gamma^0, \gamma^1 \rangle$, where $\gamma^0 = \mathbf{1}_{\ell_{000}} + \mathbf{1}_{\ell_{100}}$ and $\gamma^1 = \mathbf{1}_{\ell_{001}} + \mathbf{1}_{\ell_{011}}$.

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There is a natural isomorphism $C_i \cong C^i$ by identifying a geometric element σ with its characteristic function $\mathbf{1}_\sigma$. This induces the adjoint operator ∂^* that acts on the cochain group but behaves like the boundary operator,

$$\partial^* \mathbf{1}_\sigma = \sum_{\ell \in \partial\sigma} \mathbf{1}_\ell.$$

The Laplacian operator is defined as

$$\Delta^i = \partial^*_{i+1}\delta^i + \delta_{i-1}\partial^*_i.$$

The null space of the Laplacian operator $\ker(\Delta^i)$ consists of i-dimensional harmonic fields allowed by the underlying topology. The combinatorial Hodge decomposition renders a deep understanding of these fields

$$C^i = \mathrm{im}(\delta^{i-1}) \oplus \ker(\Delta^i) \oplus \mathrm{im}(\partial^*_{i+1}).$$

Data Availability Statement. This article has no associated data or the data will not be deposited.

Code Availability Statement. This article has no associated code or the code will not be deposited.

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